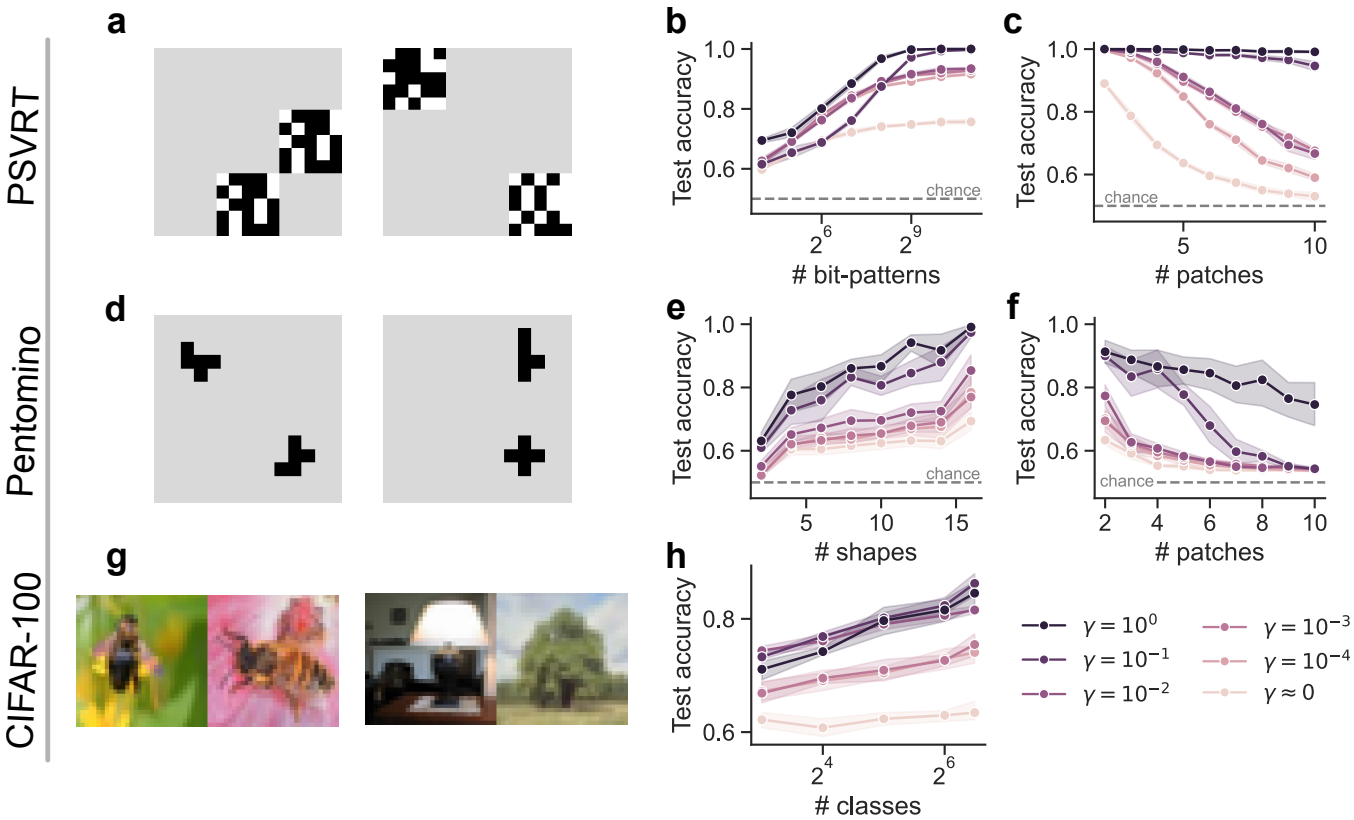

Figure 3: **Visual same-different results.** **(a)** PSVRT examples for *same* (left) and *different* (right). **(b,c)** Test accuracy on PSVRT across different numbers of training bit-patterns and image widths. Richer models learn the task with fewer patterns and exhibit less sensitivity to larger sizes. **(d)** Pentomino examples for *same* (left) and *different* (right). **(e,f)** Test accuracy on Pentomino across training shapes and image widths. As before, richer models learn the task with fewer training shapes and exhibit less sensitivity to larger sizes, though performance across models tends to diminish somewhat with increasing image size. **(g)** CIFAR-100 examples for *same* (left) and *different* (right). **(h)** Test accuracy on CIFAR-100 same-different across training classes. Richer models tend to perform better with fewer classes, though the richest model in this example performs worse. For this task, very rich models may overfit, necessitating an optimal richness level. **(all)** Shaded error regions are computed across six runs and correspond to empirical 95 percent confidence intervals.

We continue to use the same MLP model as before. Images are flattened before input to the model. Though better performance may be attained using CNNs or Vision Transformers, our ultimate goal is to study learning richness rather than maximize performance.[5] To validate our theoretical findings, we should continue to see the three hallmarks of a conceptual solution (conceptual representations, efficiency, and insensitivity to spurious perceptual details), but only in rich MLPs.

4.1 PSVRT

The parameterized-SVRT (PSVRT) dataset is a version of the Synthetic Visual Reasoning Test (SVRT), a collection of challenging visual reasoning tasks based on abstract shapes (Kim et al., 2018). PSVRT replaces the original shapes with random bit-patterns in order to better control image variability. The task input consists of an image that has two blocks of bit-patterns, placed randomly on a blank background. The model must determine whether the blocks contain the same bit pattern, or different patterns. The training set consists of a fixed number of predetermined bit patterns. The test set consists of novel bit-patterns never encountered during training.

Bit-patterns are *patch-aligned*: they occur in non-overlapping locations that tile the image. The width of an image may be specified by the number of patches. Figure 3a illustrates examples from PSVRT that are three patches wide.

Results. Figure 3b plots a model's test accuracy on PSVRT as a function of the number of training patterns. As our theory suggests, richer models learn the task more easily and generalize after substantially fewer training patterns. To test our models under perceptual variation, we consider larger image sizes. We keep the same size of bit-patterns, but increase the number of patches to make a bigger input. Figure 3c indicates that a rich model continues to perform perfectly irrespective of image size, whereas lazier models exhibit a performance de-

[5] Nonetheless, as we will soon see, an MLP performs astonishingly well on these tasks despite its simplicity — provided it remains in a rich learning regime.

cay with larger inputs.

Finally, we identify parallel/antiparallel analogs for PSVRT in the weights of a rich model (Figure A1a). The presence of these conceptual representations suggests that our theory remains a reasonable description for how a rich MLP may learn a conceptual solution to the PSVRT same-different task.

4.2 Pentomino

The Pentomino task uses inputs that are *pentomino polygons*: shapes consisting of five squares glued by edge (Gülçehre & Bengio, 2016). The input consists of an image with two pentominoes, placed arbitrarily on a blank background. The pentominoes may either be the same shape, or different. In contrast with the PSVRT task, sameness in this task implies equality up to rotation. After training on a fixed set of pentomino shapes, the model must generalize to entirely novel shapes. Like with PSVRT, shapes are patch-aligned. Figure 3d illustrates example inputs from this task that are three patches wide.

Results. Figure 3e plots a model's test accuracy on Pentomino as a function of the number of training shapes. Consistent with our theory, richer models learn the task more easily and generalize after substantially fewer training shapes. To test our models under perceptual variation, we consider larger image sizes. Like with PSVRT, we add additional patches to enlarge the input. Figure 3f indicates that a rich model continues to perform well on larger image sizes, though its performance does start to decay somewhat. Performance decays substantially faster for lazier models.

We again identify parallel/antiparallel analogs for Pentomino in the weights of a rich model (Figure A1c). The presence of these conceptual representations continues to support our theoretical perspective. Notably, Gülçehre & Bengio (2016) introduced this task to motivate curriculum learning, finding that their MLP fails to perform above chance. We found that curriculum learning is unnecessary in the presence of sufficient richness.

4.3 CIFAR-100

The CIFAR-100 dataset consists of 60 thousand real-world images, each 32 by 32 pixels (Krizhevsky & Hinton, 2009). Images belong to one of 100 different classes. In this task, the input consists of two different unlabeled images that belong either to the same or different classes. After training on images from a fixed set of labels, the model must generalize to entirely novel labels. The sets of train and test labels are disjoint, making this an extremely challenging task. The labels themselves are not provided in any form during training. Example inputs are illustrated in Figure 3g. We also experiment with providing features from VGG-16 pretrained on ImageNet (Simonyan & Zisserman, 2014). We pass CIFAR-100 images to VGG-16, then use intermediate features as inputs to our MLP. The weights of VGG-16 are fixed throughout the whole process. Note also that ImageNet is disjoint from CIFAR-100, so there is limited possibility of contamination in the test images.

Results. Figure 3h plots a model's test accuracy on CIFAR-100 images as a function of the number of training classes. We use outputs from VGG-16 block 4, layer 3, which performed the best with our model. As before, richer models tend to perform better with fewer training classes, but with a curious exception: in contrast to the previous two tasks, the richest model does not always perform decisively the best. This is particularly evident using the activations from other intermediate VGG layers, plotted in Figure G1. For certain layers and number of training classes, the optimal γ appears to be somewhat less than 1. This outcome may be in part an artifact of overfitting. Given the complexity of the task and the limited data, richer models are plausibly more susceptible to idiosyncratic features of the training set that generalize poorly, analogous to overfitting effects in classical statistics that degrade the performance of powerful models. In this case, slightly less learning richness may be the optimal setting. Since CIFAR-100 images are fixed to 32 by 32 pixels, we skipped testing variable image size for this task.

As before, we identify parallel/antiparallel analogs for this task in the weights of a rich model (Figure A1e). The general benefit of richness together with the presence of conceptual representations continues to align with our theoretical perspective. Across our three visual same-different tasks, we identified generally consistent relationships between learning richness, conceptual solutions, and good performance, supporting our theoretical findings.

5 Discussion

We studied equality reasoning using a simple same-different task. We showed that learning richness drives the development of either conceptual or perceptual behavior. Rich MLPs develop conceptual representations, learn from few training examples, and remain largely insensitive to perceptual variation. Meanwhile, lazy MLPs require exhaustive training examples and deteriorate substantially with spurious perceptual changes.

Varying learning richness recapitulates E. Wasserman et al. (2017)'s continuum between perceptual and conceptual behavior on same-different tasks. Perhaps a pigeon's competency at equality reasoning may be broadly comparable to a lazy MLP's, requiring a great deal of training and exhibiting persistent sensitivity to spurious details. Perhaps equality reasoning in human or even language-trained great apes may be comparable to a rich MLP, where learning is faster, less sensitive to spurious details, and presumably involves conceptual abstractions. We suggest that a key parameter underlying these behavioral differences may be learning richness.

Learning richness is a concept imported from machine learning theory, and it is not altogether clear how to measure richness in a living brain. Since richness specifies the degree of change in a neural network's hidden representations, the most direct analogy in the brain is to look for adaptive representations that seem to encode task-specific variables. Such approaches have implicated richness as an essential prop-

erty for context-dependent decision making, multitask cognition, generalizing knowledge, among many other phenomena (Flesch et al., 2022; Ito & Murray, 2023; Johnston & Fusi, 2023; Farrell et al., 2023). Our theory predicts that greater learning richness relates to faster generalization in equality reasoning, and look forward to possible experimental validation of this principle.

Our work also contributes to the longstanding debate on a neural network's facility with abstract reasoning. Rich MLPs demonstrate successful generalization to unseen symbols irrespective of input dimension or even high noise variance. Further, the rich MLP's development of parallel/antiparallel components suggests the formation of abstractions, supporting the account that neural networks may indeed learn to develop and manipulate symbolic representations.

Practically, we demonstrate that learning richness is a vital hyperparameter. Increasing richness generally increases test performance substantially, improves data efficiency, and reduces sensitivity to spurious details. For complex tasks tuned with a large range of γ, there may be an optimal level of richness. Indeed, for CIFAR-100, we observed that more richness is not always better, and an optimal level exists. We encourage more widespread application of richness parametrizations like μP, and advocate for adding γ to the list of tunable hyperparameters that every practitioner must consider when developing neural networks (Atanasov et al., 2024).

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References

Alhama, R. G., & Zuidema, W. (2019, 1 August). A review of computational models of basic rule learning: The neural-symbolic debate and beyond. *Psychon Bull Rev*, *26*(4), 1174–1194.

Atanasov, A., Meterez, A., Simon, J. B., & Pehlevan, C. (2024). The optimization landscape of sgd across the feature learning strength. *arXiv preprint arXiv:2410.04642*.

Battaglia, P. W., Hamrick, J. B., Bapst, V., Sanchez-Gonzalez, A., Zambaldi, V., Malinowski, M., ... Pascanu, R. (2018, 17 October). Relational inductive biases, deep learning, and graph networks. *arXiv*.

Bernstein, J., Wang, Y.-X., Azizzadenesheli, K., & Anandkumar, A. (2018). signsgd: Compressed optimisation for non-convex problems. In *International conference on machine learning* (pp. 560–569).

Biehl, M., & Schwarze, H. (1995). Learning by on-line gradient descent. *Journal of Physics A: Mathematical and general*, *28*(3), 643.

Boix-Adsera, E., Saremi, O., Abbe, E., Bengio, S., Littwin, E., & Susskind, J. (2023). When can transformers reason with abstract symbols? *arXiv preprint arXiv:2310.09753*.

Bordelon, B., & Pehlevan, C. (2022). Self-consistent dynamical field theory of kernel evolution in wide neural networks. *Advances in Neural Information Processing Systems*, *35*, 32240–32256.

Bubeck, S., Chandrasekaran, V., Eldan, R., Gehrke, J., Horvitz, E., Kamar, E., ... Zhang, Y. (2023, 13 April). Sparks of artificial general intelligence: Early experiments with GPT-4. *arXiv*.

Carstensen, A., & Frank, M. C. (2021). Do graded representations support abstract thought? *Current Opinion in Behavioral Sciences*, *37*, 90–97.

Chizat, L., & Bach, F. (2020). Implicit bias of gradient descent for wide two-layer neural networks trained with the logistic loss. In *Conference on learning theory* (pp. 1305–1338).

Chizat, L., Oyallon, E., & Bach, F. (2019). On lazy training in differentiable programming. *Advances in neural information processing systems*, *32*.

Cho, Y., & Saul, L. (2009). Kernel methods for deep learning. *Advances in neural information processing systems*, *22*.

Farrell, M., Recanatesi, S., & Shea-Brown, E. (2023, 1 December). From lazy to rich to exclusive task representations in neural networks and neural codes. *Curr. Opin. Neurobiol.*, *83*(102780), 102780.

Flesch, T., Juechems, K., Dumbalska, T., Saxe, A., & Summerfield, C. (2022, 6 April). Orthogonal representations for robust context-dependent task performance in brains and neural networks. *Neuron*, *110*(7), 1258–1270.e11.

Fleuret, F., Li, T., Dubout, C., Wampler, E. K., Yantis, S., & Geman, D. (2011). Comparing machines and humans on a visual categorization test. *Proceedings of the National Academy of Sciences*, *108*(43), 17621–17625.